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Review

Malignant transformation of oral lichen planus and oral lichenoid lesions: A meta-analysis of 20095 patient data

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ABSTRACT

Objectives: For over a century, a heated debate existed over the possibility of malignant transformation of oral lichen planus (OLP). We performed this meta-analysis to evaluate the malignant potential of OLP and oral lichenoid lesions (OLL) and investigate the possible risk factors for OLP malignant transformation into oral squamous cell carcinoma (OSCC).

Materials and methods: We searched Medline, Scopus, and Web of Knowledge for relevant observational studies. Data on OLP malignant transformation were calculated as a pooled proportion (PP), using the Der-Simonian Liard method. We performed subgroup analyses by OLP diagnostic criteria, site, and clinical type, using Open Meta[Analyst] software. Data on possible risk factors for malignant transformation were pooled as odds ratios (ORs), using Comprehensive Meta-Analysis software.

Results: Pooling data for OLP malignant transformation from 57 studies (19,676 patients) resulted in an overall PP of 1.1% [95% CI: 0.9%, 1.4%], while pooling data from 14 recent studies that used the World Health Organization-2003 diagnostic criteria resulted in an overall-PP of 0.9% [95% CI: 0.5%, 1.3%]. The risk of malignant transformation was higher (PP = 2.5%, 95% CI [1%, 4%]) in OLL patients (419 patients). A significant increase of malignant transformation risk was noted among smokers (OR = 2, 95% CI [1.25, 3.22]), alcoholics (OR = 3.52, 95% CI [1.54, 8.03]), and HCV-infected patients (OR = 5, 95% CI [1.56, 16.07]), compared to patients without these risk factors.

Conclusion: A small subset of OLP patients (1.1%) develop OSCC; therefore, regular follow-up for these patients is recommended. A higher incidence of malignant transformation was found among smokers, alcoholics, and HCV-infected patients; however, these associations should be further investigated.

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Introduction

Oral Lichen Planus (OLP) is an oral subtype of Lichen Planus, affecting 1–4% of the worldwide population with a higher frequency in middle aged and elderly women [1]. It is a chronic inflammatory disease, characterized by a T-cell mediated response against epithelial basal cells, leading to basal cell degeneration and subepithelial band like infiltration by T-lymphocytes [2,3]. Clinically, there are six different subtypes, classified into two groups: Non-erosive-atrophic forms [including reticular, papular and plaque like] and erosive-atrophic forms [including atrophic (erythematous), erosive (ulcerative) and bullous] [4].

Oral squamous cell carcinoma (OSCC) is the most common form (90%) of oral cancer, with increasing prevalence on a global scale [5]. The first report of OLP malignant transformation was published

Abbreviations: HCV, hepatitis C virus; OLL, oral lichenoid lesions; OLP, oral lichen planus; OSCC, oral squamous cell carcinoma; PP, pooled proportion.

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in 1910 by Hallopeau et al. [6]. Later, several reports were published to investigate the malignant potential of OLP with conflicting results. Several studies [7–13] have reported malignant transformation rates for OLP that ranged between 0.07% and 5.8%, while other studies [14–17] showed no malignant potential for OLP.

The major dilemma that always confused researchers regarding the malignant transformation of OLP was the lack of universal criteria for its diagnosis. In fact, Krutchkoff et al. [18] and Van Der Meij et al. [19] alerted that almost two thirds of published case reports about OLP malignant transformation were not sufficiently documented to be considered. Over decades, several efforts were exerted to overcome this obstacle. In 1978, the World Health Organization (WHO) developed diagnostic criteria for OLP that were based on clinical and histopathological standards [20]. In 2003, Van der Meij and Van der Waal suggested modifying the WHO criteria to differentiate between OLP and other lichenoid lesions, such as lichenoid dysplasia, which has a significantly higher risk of malignant transformation than OLP [21].

We performed this systematic review and meta-analysis to precisely evaluate the malignant potential of OLP and investigate the association between OLP malignant transformation and different related risk factors, such as smoking, alcohol consumption, and hepatitis C infection.

Methods

We followed the guidelines of the (MOOSE statement: meta-analysis of observational studies in epidemiology) during the preparation of this meta-analysis [22].

Literature search strategy

We searched Medline (through PubMed), Scopus, and Web of Knowledge, using the following search strategy [Oral lichen planus OLP OR Oral lichinoid lesion OR OLL AND Malignent OR Cancer OR Oral squamous cell carcinoma OR OSCC]. No restrictions by publication period were used. We also scanned the bibliography of included studies to identify any missed studies that are relevant to our subject.

Eligibility criteria and study selection

We included full length articles of observational studies, published in the English language, in peer reviewed medical journals. All observational studies, providing dichotomous data on the malignant potential of OLP or oral lichenoid lesions (OLL), were included. We excluded conference abstracts, articles published as a whole in a non-English language, case reports/series, and secondary analysis articles. Two independent reviewers applied the selection criteria in two steps [Title and abstract screening, then full text screening] to determine the included studies and disagreements were solved upon the opinion of a third reviewer.

Data extraction

Two independent reviewers extracted the following data from included studies: (I) Characters of study design and length of follow up period, (II) risk of bias assessment domains, (III) characteristics of OLP lesions including the used diagnostic system criteria, site, and clinical type [Atrophic, erosive, reticular, or plaque], (IV) the incidence of malignant transformation in OLP or OLL cases, and (V) the incidence of malignant transformation in different subgroups of OLP patients. Disagreements were resolved upon the opinion of a third reviewer.

Risk of bias assessment

We used the Newcastle-Ottawa scale [23] to assess the risk of bias within the included observational studies. Using this scale, each included study is evaluated based on reporting three essential domains:

(a) Selection of the study subjects i.e. included patients were derived from a representative population, and the exposure (OLP diagnosis) was ascertained based on predefined criteria (preferably including histological examination).
(b) Comparability of groups on demographic characteristics and important potential confounders: which is achieved by adequate control of secondary risk factors, such as smoking and alcohol consumption, as well as adequate separation between OLP and OLL cases (by excluding history of medications or restorations).
(c) Ascertainment of the prespecified outcome (exposure/treatment): based on histological assessment with a sufficient follow up duration.

Due to the design of included studies (lack of non-exposed group), we removed the question about the selection of the non-exposed group (Question No. 2) from the Newcastle-Ottawa Scale; therefore, the total score is out of eight, not nine [24]. Each article was evaluated by two independent reviewers and any disagreements were resolved by consenesus among both reviewers. We assessed for the existence of publication bias, using the Egger's test [25].

Data synthesis

Pooled proportion (PP) of the overall risk of malignant transformation in OLP and OLL was calculated under the random effects model, using Der-Simonian Liard method. We conducted a cumulative meta-analysis to display the trend of malignant transformation over time from 1924 to 2016. Subgroup analysis was conducted to compare the risk of malignant transformation according to the diagnostic criteria, clinical type, and site of OLP. We calculated a pooled odds ratio (OR), along with its 95% confidence interval (CI), to compare the risk of transformation according to patients' gender, smoking, alcohol consumption, infection with hepatitis C virus (HCV), and diabetes mellitus. Comprehensive Meta-Analysis and Open Meta [Analyst] software were used for quantitative data synthesis.

Results

Literature search results

Literature search retrieved 1200 unique records. After title/abstract screening, 197 full text articles were retrieved for further evaluation. Finally, 57 studies that provided data on 19,676 OLP and 419 OLL cases were eligible for inclusion in our analysis (Fig. 1). Of OLP cases, a total of 280 patients developed OSCC. The rate of malignant transformation of OLP in individual studies ranged from 0.3% to 14.3%. Baseline data of enrolled patients in included studies are illustrated in Tables 1. Table 2 displays a comparison between demographic characteristics of OSCC patients in included studies and the reference profile for OSCC patients (from the Surveillance, Epidemiology, and End Results [SEER] database).

Risk of bias assessment

The majority of included studies reported adequately on the representativeness of their cohorts (55 of 57), exposure

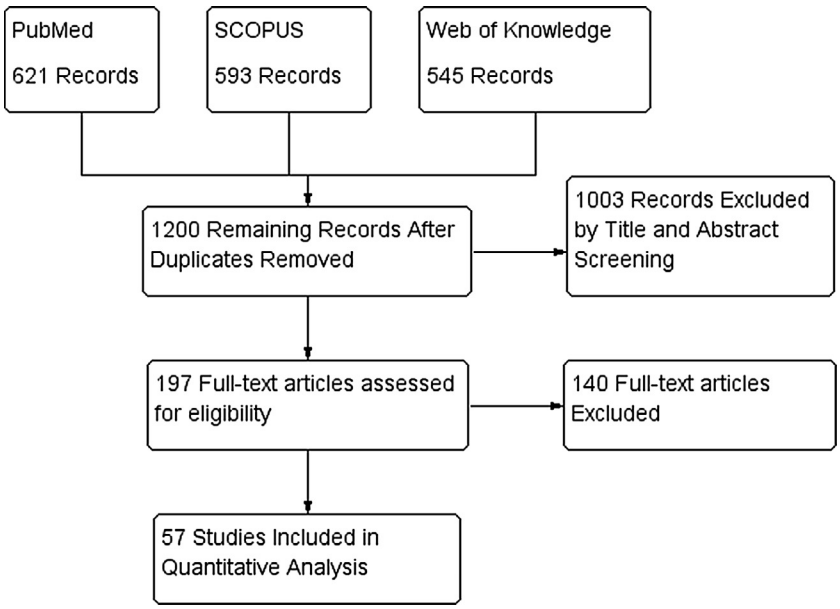

Fig. 1. A flow diagram of literature search and study selection process.

ascertainment (46 of 57), and outcome assessment methods (50 of 57). However, a lower number of studies reported on the length of (29 of 57) and loss during follow up (13 of 57). Control of secondary risk factors and similar lesions were reported in 37 and 36 studies, respectively. Details for risk of bias assessment are illustrated in Supplementary file 1.

Meta-analysis for risk of malignant transformation

Single-arm meta-analysis, under the random effects model and using Der-Simonian Liard method, resulted in an overall pooled proportion (PP) of 1.1% [95% CI: 0.9–1.4%]. There was a considerable heterogeneity among the results of individual studies, as indicated by an I^2 value of 60.96% and Chi-Square $p < 0.001$ (Fig. 2A). A cumulative analysis indicated a decreasing trend that started by 5.9% transformation rate in 1924 and went down to 1.5% in 1968. From 1972 to the present time, there was a relatively consistent rate that ranged between 0.5% and 1.1% (Fig. 2B). Considering the significant heterogeneity, we performed a sensitivity analysis to measure the influence of omitting single studies on the overall PP. The summary PP was consistent on removing any single study and ranged between 1.1% and 1.2% (Supplementary file 2).

We performed a subgroup analysis for the recent studies in which OLP was defined according to the 2003 modified WHO criteria [21]. The PP was 0.9% with a 95% CI ranging from 0.5% to 1.3%. A moderate degree of heterogeneity was identified among these studies ($I^2 = 34.84\%$, $p = 0.096$) (Fig. 3A). Publication bias towards studies with transformation rate less than 1.1% was evident by Egger's regression test ($p = 0.00038$) and asymmetry of the funnel plot (Fig. 3B).

Data on the overall malignant transformation of OLL into OSCC were provided by four studies [15,59,67,68] that included 419 OLL cases. Of them, 13 cases were complicated by malignancy with a rate of transformation in individual datasets, ranging between 1.2% and 4.9%. The PP of malignant transformation of OLL was 2.5% [95% CI: 1–4%], with no heterogeneity among pooled studies ($I^2 = 0\%$, $p = 0.47$).

Subgroup analysis

Stratification according to the clinical type of OLP indicated that the rate of malignant transformation was 1.7% in erosive OLP, 1.3% in atrophic OLP, and 0.1% in the reticular type (Fig. 4). The most common site of transformation was the tongue (1.05%), followed by the buccal mucosa (0.7%). The gingiva and lips had similar rates of malignant transformation (0.6%), while the floor of the mouth developed OSCC at a rate of 0.5%.

Risk factors for OLP malignant transformation

Twenty-eight studies provided data on rates of transformation among males and females. Malignant transformation was more prevalent among males than females (OR = 1.11); however, this difference was not statistically significant ([95% CI: 0.83, 1.48], $p = 0.48$) (Fig. 5A).

Compared to non-smokers, smokers had a significantly higher rate of malignant transformation (OR = 2, 95% CI [1.25, 3.22], $p = 0.004$). Twenty-six patients among the 1268 smokers developed malignancy, compared to 49 of 4387 non-smokers. These data were reported by 13 homogenous studies ($I^2 = 0\%$, $p = 0.68$) (Fig. 5B). Similarly, pooled data from five studies showed increased rates of malignancy among alcoholics compared to non-alcoholics (OR = 3.52, 95% CI [1.54, 8.03], $p = 0.003$) (Fig. 5C).

Interestingly, data from two studies (561 OLP cases) displayed an increased prevalence of malignancy among HCV +ve patients in comparison to the HCV −ve group (OR = 5, 95% CI [1.56, 16.07], $p = 0.007$) (Fig. 5D). On the other hand, there was no significant difference between diabetic and non-diabetic patients (OR = 1.49, 95% CI [0.48, 4.62], $p = 0.49$) (Fig. 5E).

Discussion

Our meta-analysis of 57 studies (20,095 OLP cases) showed that 1.1% of OLP patients developed OSCC, while the rate of malignant transformation among OLL cases was 2.5%. Selective pooling of studies that used the WHO 2003 diagnostic criteria showed a malignant transformation rate of 0.9%, which was slightly lower than the overall estimate. Although the majority of included studies were heterogenous in terms of design, inclusion criteria, and length of follow up [which was also expressed in our analysis: $I^2 = 60.96\%$], most of them reported a malignant transformation rate of nearly 1%, which is consistent with our findings.

Table 1
shows baseline data of enrolled OLP patients in included studies.

Study ID	Country of origin	Total cases of OLP	Subtypes of OLP		Cases Of SCC	Malignant transformation			Mean follow up time (months)	Criteria used for diagnosis of OLP	Risk factors in SCC cases			
			Red	White		Mean time to transformation (months)	Rate (%)	Annual Rate (%)			HCV	Alcohol	Smoking	Diabetes
Gonzalez-Moles et al. [69]	Spain	21	–	–	3	9.72	14.3	–	–	Described	–	–	–	–
Irani et al. [61]	Iran	112	60 (53.6)	52 (46.4)	1	36	0.008	–	–	Modified WHO 2003	–	–	–	–
Lauritano et al. [64]	Italy	87	34 (39)	53 (61)	1	60	1.1	–	60	Modified WHO 2003	0	0	0	0
Nuzzolo et al. [63]	Italy	13	10 (77)	3 (23)	1	–	7.7	–	61.8	Described	0	0	0	0
Varghese et al. [62]	India	122	41 (33.6)	81 (66.4)	2	–	5	–	–	Modified WHO 2003	–	–	–	–
Richter et al. [70]	Croatia	563	153 (27.2)	410 (72.8)	4	91.2	0.7	–	90	WHO diagnosis system-ICD9	–	–	3	–
Casparis et al. [59]	Switzerland	381	–	–	5	18.96	1.3	–	–	Described	–	0	2	2
Radochová et al. [17]	Czech	171	92 (53.8)	79 (46.2)	0	0	0	–	–	Modified WHO 2003	0	0	0	0
Wang et al. [60]	Taiwan	381	–	–	2	8.07	0.52	–	–	Described	–	–	–	–
Bardellini et al. [57]	Italy	204	56 (27.5)	148 (72.5)	2	50	0.98	–	43.2	Modified WHO 2003	2	–	0	–
Gümrü [58]	Turkey	370	224 (60.5)	146 (39.5)	1	24	0.27	–	–	Modified WHO 2003	0	0	0	0
Kaplan et al. [10]	Israel	171	113 (66.1)	58 (33.9)	6	–	3.5	–	51.6	Modified WHO 2003	–	–	–	–
Shen et al. [13]	China	518	247 (47.7)	271 (52.3)	5	70.4	0.96	–	40.5	WHO, 1978	–	0	0	–
Bombeccari et al. [56]	Italy	327	–	–	8	39.3	2.4	0.36	81.7	Modified WHO 2003	0	0	0	2
Bermejo-Fenoll et al. [11]	Spain	550	359 (65.3)	191 (34.7)	5	–	0.9	–	–	WHO, 1978	–	0	0	–
Oliveira et al. [16]	Brazil	110	56 (51)	54 (49)	0	–	0	–	–	Described	0	0	0	0
Torrente-Castells et al. [12]	Spain	65	22 (33.8)	43 (66.2)	2	–	3.1	–	–	Modified WHO 2003	–	2	2	–
Carbone et al. [8]	Italy	808	332 (41.1)	476 (58.9)	15	52.33	1.86	0.69	43.2	Modified WHO 2003	–	3	5	1
Pakfetrat et al. [9]	Iran	420	129 (30.7)	291 (69.3)	3	26	0.07	–	–	Described	–	0	0	–
Hsue et al. [72]	Taiwan	143	–	–	3	14.7	2.1	–	42.64	Described	–	3	3	–
van der Meij et al. [15]	Netherland	67	–	–	0	–	0	–	55.9	Modified WHO 2003	0	0	0	0
Bornstein et al. [55]	Sweden	141	–	–	1	–	0.71	–	–	Not Described	1	–	1	1
Ingafou et al. [7]	UK	690	165 (23.9)	525 (76.1)	13	84 (Median)	1.9	–	–	Described	–	–	–	–
Laeijendecker et al. [66]	Netherland	200	108 (54)	92 (46)	3	51.6	1.5	–	120 51.6	Described	–	–	1	1
Xue et al. [54]	China	674	328 (48.6)	346 (51.4)	4	108	0.6	–	–	Modified WHO 2003	–	3	3	0
Gandolfo et al. [53]	Italy	402	240 (59.7)	162 (40.3)	9	44.56	2.2	–	58.8	Described	4	2	4	1
Rodstrom et al. [52]	Sweden	1028	–	–	5	77.76	0.5	>0.1	76.8	WHO diagnosis system-ICD9	–	–	–	–
Lanfranchi et al. [51]	Argentina	491	308 (62.7)	183 (37.3)	32	20.8	6.51	–	–	Described	–	–	–	–
Van der Meij et al. [68]	The Netherlands	62	–	–	0	–	0	–	31.9	Modified WHO 2003	–	–	–	–
Eisen [50]	USA	723	456 (63)	267 (37)	6	41	0.8	–	54	Described	–	0	0	–
Chainani-Wu et al. [49]	USA	229	173 (75.5)	56 (24.5)	4	168	1.7	–	–	Described	–	–	–	–
Mignogna et al. [48]	Italy	502	–	–	21	–	3.7	–	–	Described	5	–	–	–
Rajentheran et al. [47]	UK	832	–	–	7	72	0.8	–	132	Krutchkoff et al. [18]	–	1	–	–
Muzio et al. [46]	Italy	263	–	–	14	84	5.32%	–	68.4	Described	3	0	3	2
Markopoulos et al. [45]	Greece	326	115 (31.8)	211 (68.2)	4	78	1.3	–	57.6	Described	0	0	–	–

(*continued on next page*)

Table 1 (*continued*)

Study ID	Country of origin	Total cases of OLP	Subtypes of OLP		Cases Of SCC	Malignant transformation			Mean follow up time (months)	Criteria used for diagnosis of OLP	Risk factors in SCC cases			
			Red	White		Mean time to transformation (months)	Rate (%)	Annual Rate (%)			HCV	Alcohol	Smoking	Diabetes
Gorsky et al. [44]	Israel	157	90 (57.3)	67 (42.7)	2	58.5	1.3	–	–	Described	0	0	–	–
Barnard et al. [42]	UK	241	–	–	9	–	3.7	–	–	Described	–	–	4	1
Brown et al. [43]	USA	193	124 (64.2)	69 (35.8)	0	–	0	–	–	Described	0	0	–	–
Monoarz et al. [65]	Israel	280	–	–	8	57	2.85	–	–	Krutchkoff et al. [18]	–	0	3	0
Voute et al. [41]	The Netherlands	113	–	–	3	48	2.6	–	84	Described	–	–	–	–
Sigurgeirsson and Lindelof [71]	Sweden	2071	–	–	8	55.5	0.4	–	–	Described	–	–	–	–
Silverman et al. [40]	USA	214	153 (71.5)	61 (29.5)	5	108	2.3	–	–	Described	–	–	–	–
Vincent et al. [39]	USA	100	79 (79)	21 (21)	0	–	0	–	9.1	Described	0	0	0	0
Salem [38]	Saudi Arabia	72	50 (69.4)	22 (31.6)	4	–	5.6	–	38.4	Described	–	–	–	–
Holmstrup et Al [37]	Denmark	611	–	–	9	121.2	1.5	0.2	90	Described	–	0	4	0
Murti et al. [36]	India	722	–	–	3	92	0.4	0.083	61.2	Described	–	–	3	–
Silverman et al. [35]	USA	570	389 (68.2)	181 (31.8)	7	40.8	1.2	–	66	Described	–	–	2	–
Silverman and Griffith [34]	USA	200	119 (59.5)	81 (41.5)	5	–	2.5	–	–	Described	–	–	–	–
Kovesi and Banoczy [32]	Hungary	326	193 (59.2)	133 (40.2)	1	–	0.4	–	36	Described	–	–	–	–
Fulling [33]	Denmark	225	–	–	1	–	0.4	–	43.2	Described	–	–	–	–
Shklar [31]	USA	600	–	–	3	–	0.5	–	–	Described	–	–	–	–
Andreasen [29]	Denmark	115	49 (42.6)	66 (57.4)	0	0	0	–	31.2	Described	0	0	0	0
Cawson [30]	UK	138	–	–	1	–	0.7	–	–	Described	–	–	–	–
Altman and Perry [28]	USA	127	–	–	1	–	0.8	–	–	Described	–	–	–	–
Warin et al. [27]	UK	53	–	–	4	–	9.0	–	–	Described	–	–	1	–
Warin and Crabb [69]	UK	45	–	–	2	(Range) 120–156	0.04	–	–	Described	–	–	–	–
Montgomery and Culver [26]	UK	17	–	–	1	–	6.0	–	–	Described	–	–	–	–

Abbreviations: HCV: Hepatitis C virus, OLP: Oral lichen planus, SCC: Squamous cell carcinoma, WHO: World Health Organization.

Table 2
shows a comparison of baseline demographic data between squamous cell carcinoma (SCC) cases in our analysis and reference statistics for SCC in the oral cavity.

		Cases of oral squamous cell carcinoma in our analysis	Reference statistics for oral cancer (SEER database 2012–2013 data)
Lifetime risk		1.1%	1.1%
Average age Ys (Mean ± SD)		58.49 ± 8.010	62
Gender N (%) [28 studies]	Male	69 (35.4%)	70%
	Female	126 (64.6%)	30%
Oral location %	Tongue	30.4% (12 studies)	40%
	Buccal mucosa	20.3% (12 studies)	–
	Gingiva	17.4% (12 studies)	35%
	Mouth floor	14.5% (11 studies)	17%
	Lips	17.4% (8 studies)	–
	Others	–	8%

N.B. The reported demographic data were derived from a subset of included studies that clearly indicated such data. The number of studies that reported on each item is clarified beside it.

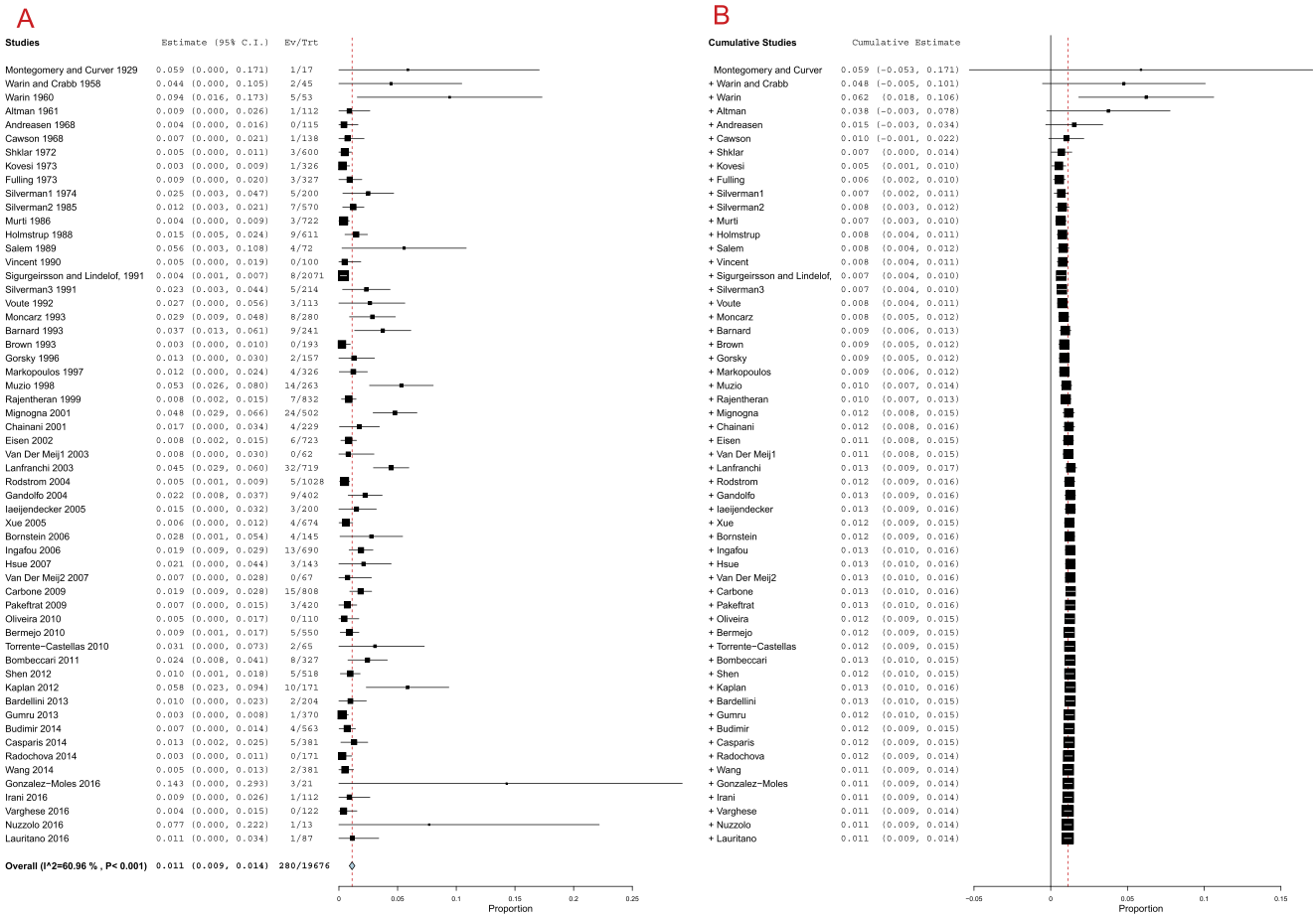

Fig. 2. Forest plots for (A) the pooled proportion of OLP malignant transformation under the random-effects model, using Der-Simonian Liard method, and (B) cumulative meta-analysis showing the trends of OLP malignant transformation from 1924 till 2016.

Several theories were developed to explain the possible malignant transformation of OLP. A widely-established hypothesis is the production of nitric oxide (NO) by inflammatory cells within the lesion [73], which later reacts with O_2 to induce the formation of 8-dihydro-2¢-doxyguanosine (8-oxodG) [74]. This compound is known to cause G-T nucleotide transversion, promoting carcinogenesis [75]. Another possible source of DNA mutation is the activity of cyclo-oxygenase II enzyme within the infiltrating inflammatory cells, increasing the release of inflammatory cytokines and malondialdehyde, a carcinogenic metabolite that is known to cause DNA damage [76]. Multiple studies have reported a high proliferation rate of the basal epithelial cells within OLP lesions [77–79] and it was shown that this rate is higher in patients who later developed OSCC than those who did not [78,80]. This phenomenon was explained later by the deficient function of p53 apoptotic protein and overactivation of the Akt/Protein kinase-B cell survival pathway [79,81,82].

The relationship between tobacco smoking and malignant transformation of OLP is highly debatable. Established as a carcinogen, the main argument is whether smoking can independently cause OSCC or just interacts with OLP to increase its malignant potential. In the modified WHO (2003) criteria, smoking must be properly documented to differentiate true malignant transformation from cases of conventional OSCC that happened to occur in

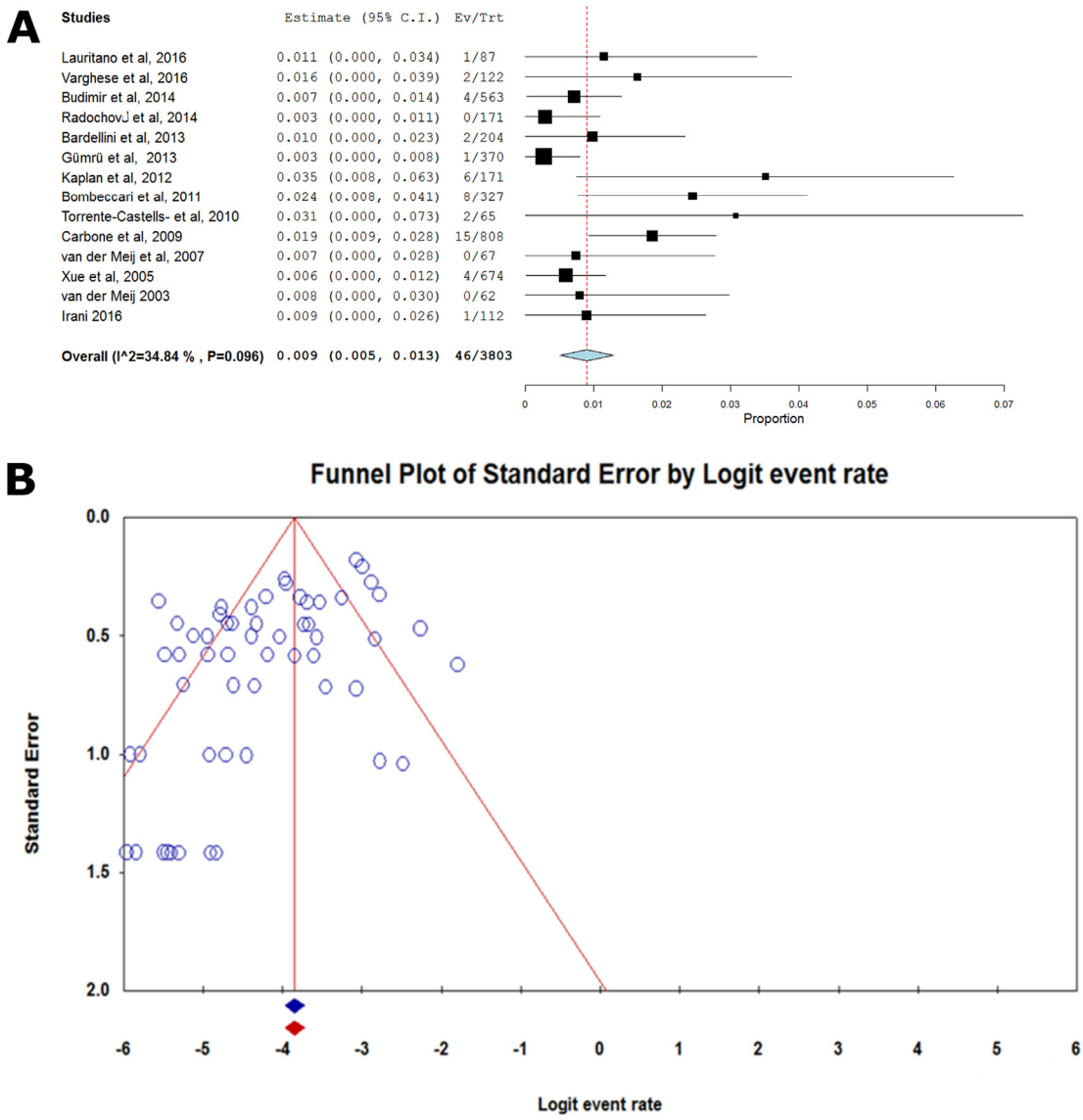

Fig. 3. (A) forest plot for subgroup analysis for included studies that used the 2003 modified diagnostic criteria of the World Health Organization, and (B) funnel plot of publication bias for all included studies.

OLP patients [21]. According to our analysis, tobacco smoking increases the risk of OLP malignant transformation; however, future studies should consider the points mentioned earlier.

Another point of controversy is the relationship between hepatitis C virus and OLP malignant transformation. Few studies found a high OLP malignant transformation rate in HCV endemic areas and HCV-RNA strands were detected in OLP tissues [83–85]. However, OSCC cases were previously reported in HCV-infected patients, irrespective of OLP [84,86]. Of note, HCV can induce carcinogenesis directly by deregulating the cell cycle [87,88] or indirectly by causing cirrhosis [89]. Our meta-analysis showed a significant association between HCV infection and OLP malignant transformation; however, this finding is based on pooling of two studies; therefore, it needs further confirmation and validation.

Strength points

We performed a comprehensive, yet careful selection of included studies to provide the most robust evidence about the malignant transformation of OLP. Our review is the first to investigate the potential of gender, smoking, alcohol consumption, and HCV as risk factors for OLP malignant transformation in a meta-analysis framework. Due to the lack of a control group from the general population, we compared our data to the reference profile for OSCC in the SEER database.

We tried to overcome the possible diagnostic confusion between OLP and other lichenoid lesions in our meta-analysis by (1) performing a subgroup analysis of the studies that exclusively used the modified WHO 2003 or matching criteria, (2) separating the cases of OLP and OLL in different analytic datasets to calculate the malignant transformation of each pathological condition independently, and (3) conducting a sensitivity analysis to ensure that none of the individual studies, in which diagnostic confusion could have occurred, significantly shifted the overall effect estimate in its direction.

Limitations

We only included studies, published in the English language; however, there is an evidence that restricting a meta-analysis to

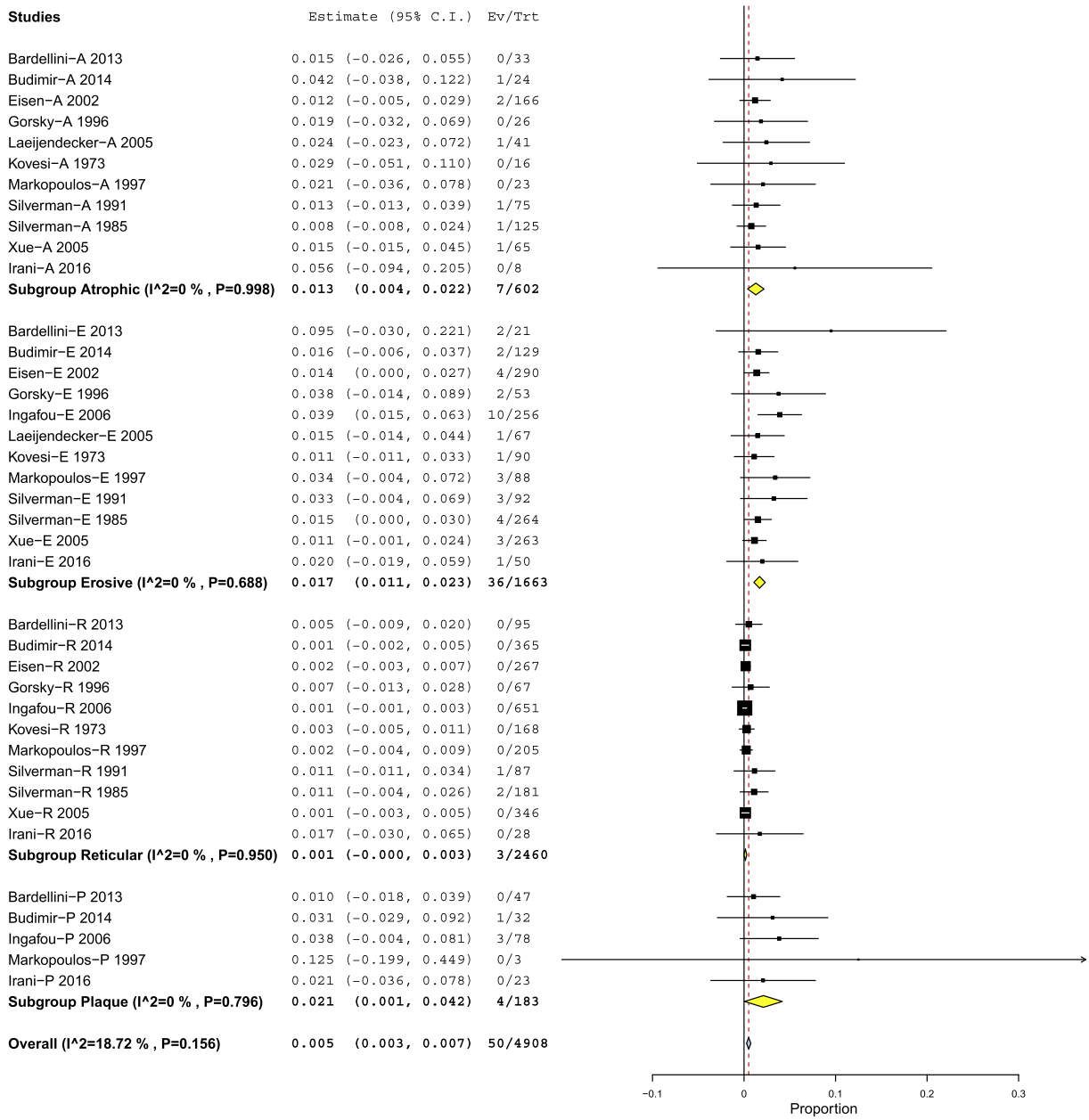

Fig. 4. A forest plot of stratification analysis by the clinical type of OLP, displaying an increased risk of malignant transformation among plaque and erosive types.

English studies does not introduce bias to its results [90]. Despite our efforts, it may not be fully possible to discriminate OLP from other lichenoid lesions in included studies. Moreover, the site of OSCC was not reliably reported in the majority of included studies, opening the possibility that some OSCC cases have arisen in previously-healthy oral areas. There are other risk factors that were not investigated in this analysis due to the lack of numerical data, such as the role of infective agents (Candida and human papilloma virus, in particular) and immunosuppressive drugs.

Recommendations for future research

Based on the nature of the condition, the best study design to investigate the malignant potential of OLP would be multicenter cohort studies with long follow up periods. Proper clinical and histopathological documentation and careful differentiation between OLP and other lichenoid lesions should be a priority in future studies. To provide more uniform and comparable data, we encourage investigators to use the same diagnostic criteria that can clearly define OLP lesions and eliminate confounding risk factors, such as smoking and alcohol consumption. Further studies are required to investigate the relationship between OLP malignant transformation and risk factors, such as smoking, immunosuppressive drugs, and infections (especially candida, HPV, and HCV).

Implications for clinical practice

Based on our findings, we recommend regular follow up of OLP patients to ensure early cancer detection. There is a debate in the literature over the optimal number of annual visits; however, two visits per year seems adequate and technically feasible [15]. Moreover, Muzio et al. found no significant value in terms of mortality with a higher frequency of annual visits [46]. Regarding the healthcare professionals who should perform the follow up, Gan-

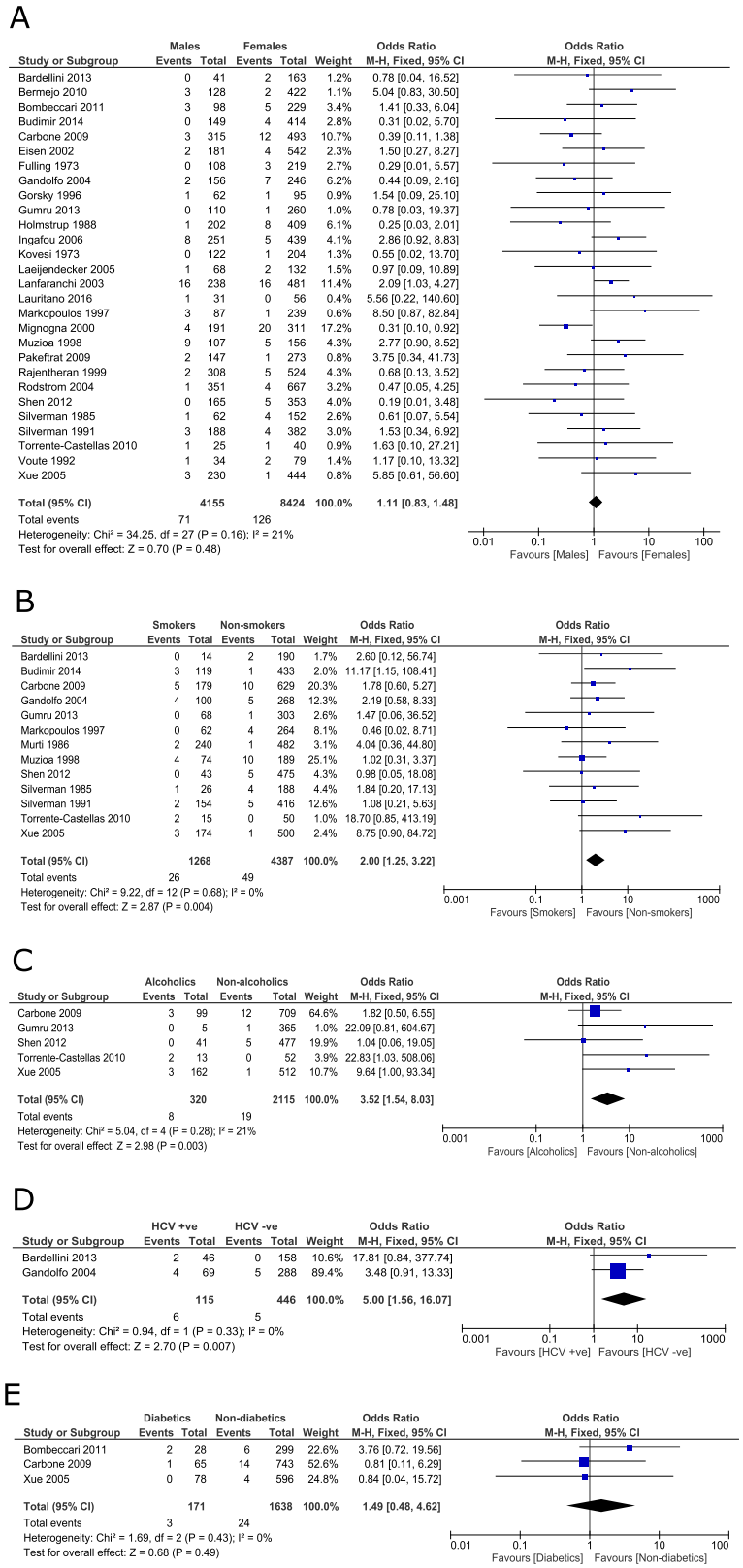

Fig. 5. Forest plots for odds ratios of the association between malignant transformation and (A) gender, (B) smoking, (C) alcohol consumption, (D) hepatitis C infection, and (E) diabetes mellitus.

dolfo et al. suggested it should be performed by general dentists to enhance their training in early cancer detection [53]. Moreover, there is no evidence in the literature that supports the need for oral medicine specialists to perform the follow-up examinations.

The premalignant potential of OLP, the possible risk factors for such transformation, and the importance of follow up should be communicated to OLP patients. Until a high-quality evidence is generated regarding the risk of treating OLP lesions with immuno-

supressive drugs on malignant transformation, these drugs should be allowed in the clinical management of OLP lesions.

Conclusion

Our meta-analysis supports the notion that a small subset of OLP patients (1.1%) develop OSCC. A higher incidence of malignant transformation was found among smokers, alcohol consumers, and HCV-infected patients; however, these associations should be interpreted with caution. Therefore, we recommend regular follow up of OLP patients to ensure early cancer detection. Future studies should follow a uniform criteria for OLP diagnosis and investigate the effect of smoking, infections, and immunosupressive drugs on the potential of OLP malignant transformation.

Conflict of interest

None.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.oraloncology.2017.03.012.

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